



Airtel IQ Telephony Integration with VoiceBOT using WebSocket Streaming Integration Guide

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1. Introduction

This document provides a high-level and implementation-oriented guide for integrating an external Partner VoiceBOT Platform with the Airtel IQ CPaaS Platform. It is intended for AI VoiceBOT vendors, solution architects, and engineering teams who will design, build, certify, and operate a VoiceBOT integrated with Airtel IQ.

It outlines the reference architecture, ownership boundaries, supported integration approaches, the WebSocket media-streaming contract (every message exchanged between Airtel IQ and your server), and the onboarding/certification process required to take an integration into production.

2. Airtel IQ CPaaS & Partner Integration Overview

Airtel IQ is a telecom-grade Communications Platform as a Service (CPaaS) offering from Airtel. It exposes programmable voice, real-time media streaming, and intelligent call control APIs to enable enterprise voice automation, conversational AI, and contact-center modernization.

Airtel IQ separates call control and media streaming to achieve low latency, high scalability, and clear ownership boundaries between telecom infrastructure (Airtel) and the AI Platform (Partner).

2.1 Logical Components

The Airtel IQ CPaaS platform is composed of the following logical components:

- PSTN / SIP Network
- Airtel IQ Voice Gateway (Call Flows and APIs)
- Airtel IQ Media Stream Controller
- Partner VoiceBOT Platform (Orchestration, STT, NLU/LLM, TTS)
- Contact Center / Human Agent Platform (Optional)

2.2 Use Cases

Voice Bot platform partners can integrate their conversational AI solution with Airtel IQ telephony and WebSocket interfaces. The integration enables real-time handling of inbound and outbound voice interactions by connecting Airtel IQ call control and media streaming capabilities with partner-hosted AI engines for speech recognition, natural language processing, and text-to-speech orchestration. Through this integration, enterprises can deploy AI-powered voice bots for customer support, lead qualification, appointment management, collections, authentication, and other automated voice workflows while leveraging Airtel IQ's carrier-grade telephony infrastructure, call routing, and scalability.

2.3 High Level Architecture

The diagram below summarizes the end-to-end reference architecture for Airtel IQ CPaaS integrated with a Partner VoiceBOT platform and a Contact Centre for Human Agent fallback (optional).

Airtel IQ CPaaS — VoiceBOT Partner & Agent Reference Architecture

Call control & media termination owned by Airtel IQ | ASR / NLU / TTS bot logic owned by Partner

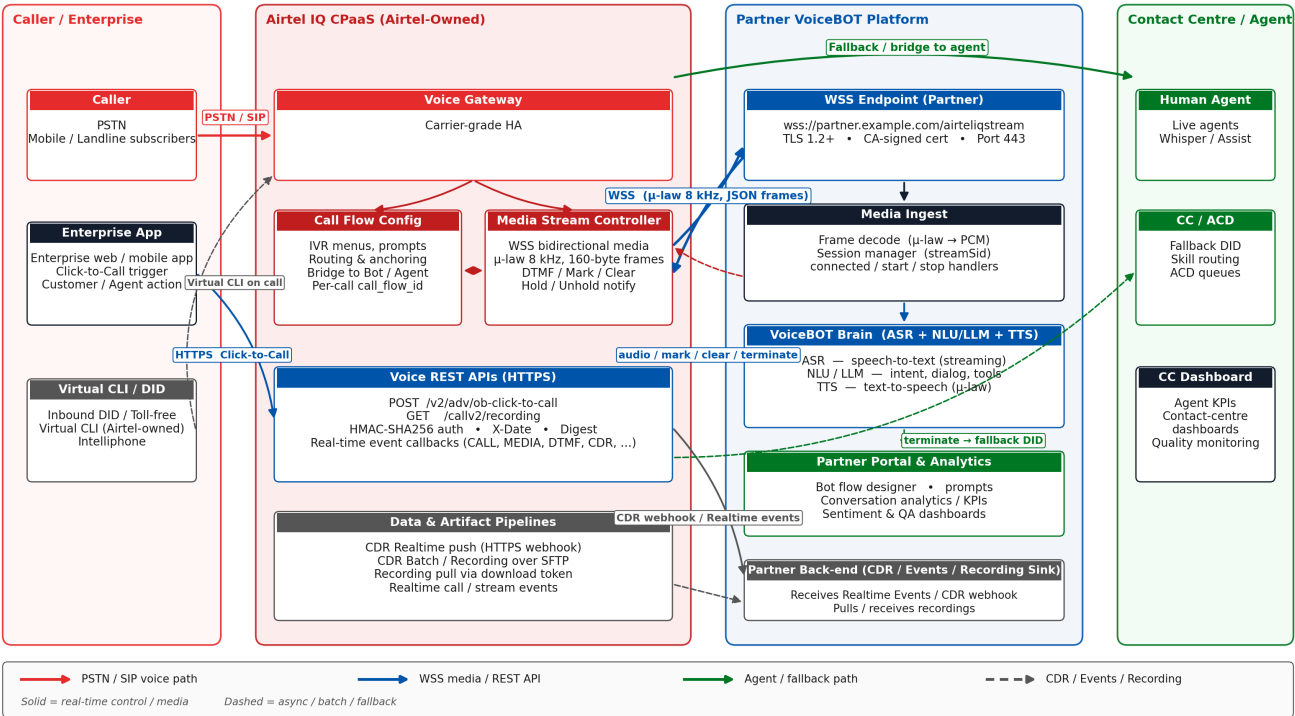


Figure 1 — Airtel IQ CPaaS, VoiceBOT Partner & Agent Reference Architecture

3. Ownership Summary (Airtel vs Partner)

The following matrix outlines the division of responsibilities between Airtel IQ — which provisions the underlying CPaaS, media streaming, and telecom infrastructure — and the Partner, who governs the Voice AI application, conversational logic, and data processing.

Area	Asset	Owner
CPaaS (Telecom)	Number (DID / Virtual CLI / Intelliphone)	Airtel IQ
	Telephone / SIP Trunk / PSTN Termination	Airtel IQ
	Call Flow Configuration (Routing / Anchoring)	Airtel IQ
	Call / Campaign Performance Dashboard	Airtel (consumed by Partner)
	CDR / Recording / Realtime Events Delivery	Airtel IQ (delivers); Partner/Customer (retains)
	Media Streaming Infrastructure (WebSocket Gateway)	Airtel IQ
Voice AI Platform	Telecom Regulatory Compliance (TRAI Guidelines)	Airtel IQ
	VoiceBOT Platform (ASR, NLU/LLM, TTS, BOT Logic)	Partner
	VoiceBOT Orchestration (Flow Design)	Partner
	API Gateway / WSS Endpoint Hosting	Partner
	Knowledge Base, Agent Features, Tools, Integrations, MCPs	Partner
	Voice / Conversation Analytics Dashboard	Partner

Area	Asset	Owner
	Monitoring, Metering, Disposition & Analytics Dashboard	Partner
	Custom Voice Cloning / Branding	Partner
Shared	End-to-End Latency Optimization	Shared (Airtel IQ & Partner)
	UAT (User Acceptance Testing) & Integration Certification	Shared (Airtel IQ & Partner)
	Incident Management & L2/L3 Technical Support	Shared (Airtel IQ & Partner)

4. Integration Approach – IQ & Partner

The integration supports Inbound, Outbound, Click-to-Call (C2C), Campaign, and Dashboard experiences. All call control and call-flow configuration remain Airtel-controlled; the Partner plugs in via their Voice Agent Platform over WebSocket using the Airtel IQ Media Streams contract.

4.1 Telephony Integration

Airtel IQ supports multiple telephony interaction modes. In each case, Airtel IQ anchors the call, manages PSTN/SIP connectivity, and bridges media to the Partner VoiceBOT Platform over a secure WebSocket connection.

4.1.1 Inbound

- Virtual CLI (Airtel-owned number) receives the incoming call.
- Call Flow Configuration (Airtel-controlled) anchors the call at Airtel IQ and bridges to the Partner VoiceBOT over WSS.
- Media is streamed to the Partner VoiceBOT over WSS in real time.
- Optional hand-off / transfer to Human Agent is supported.

4.1.2 Click-to-Call (C2C) Campaign (API)

- Customer or agent application triggers a call via the Airtel IQ C2C API.
- Airtel IQ originates both legs (Customer & VoiceBOT) and bridges them.
- Media is streamed to the Partner VoiceBOT over WSS.
- For API/Dashboard-driven campaigns: Call Flow Configuration (Airtel-controlled) anchors and bridges to the Partner VoiceBOT over WSS.
- Optional hand-off / transfer to Human Agent is supported.

NOTE: Realtime CDRs Push & Realtime Events Push are not supported for High Volume Campaigns. SFTP Push-based delivery is performed on a customized period (e.g. D+1).

4.1.3 Dashboard

Airtel IQ Dashboard

- Call/Campaign Performance, Number Lifecycle, Call Flow & Template Configuration, CDR, Recording Download.

Partner Dashboard

- VoiceBOT Agents Flow/Lifecycle Design (Workflow, Instructions, Dispositions, Tools, Knowledge Base etc.), Conversation Analytics, VoiceBOT KPIs, Associated Metering & Monitoring.

Customer / Call-Center Dashboard

- IVR-ACD Configuration, Call Flow/Lifecycle Design, Human Agent Performance, Contact-Center KPIs, etc.

4.2 Data & Artifact Pipeline Integration

Airtel IQ provides three distinct delivery pipelines to partners: a recording pipeline, a CDR pipeline, and a real-time events pipeline — ensuring timely access to call data, recordings, and call-lifecycle events.

4.2.1 Recording Pipeline

Airtel IQ offers call recording with two delivery methods:

- Pull Mechanism: Retrieve recordings via a download link provided within the Call Detail Records (CDRs).
- Push Mechanism: Automatically transfer recordings to a partner or customer endpoint using SFTP.

4.2.2 CDR Pipeline

CDR data can be delivered using the following push mechanisms:

- Batch Push Mechanism: Periodic transfers to a designated partner or customer endpoint via SFTP.
- Near Real-Time Push Mechanism: Immediate delivery of CDRs directly to a customer-hosted HTTPS endpoint.

NOTE: Realtime CDRs Push & Realtime Events Push are not supported for High Volume Campaigns. SFTP Push-based delivery is performed on a customized period (e.g. D+1).

4.2.3 Realtime Events Pipeline

Call lifecycle events can be delivered using the following push mechanism:

- Real-Time Push Mechanism: Immediate delivery of events directly to a customer-hosted HTTPS endpoint.

NOTE: Realtime CDRs Push & Realtime Events Push are not supported for High Volume Campaigns.

4.3 Media Streaming based Voice Agent Platform Integration

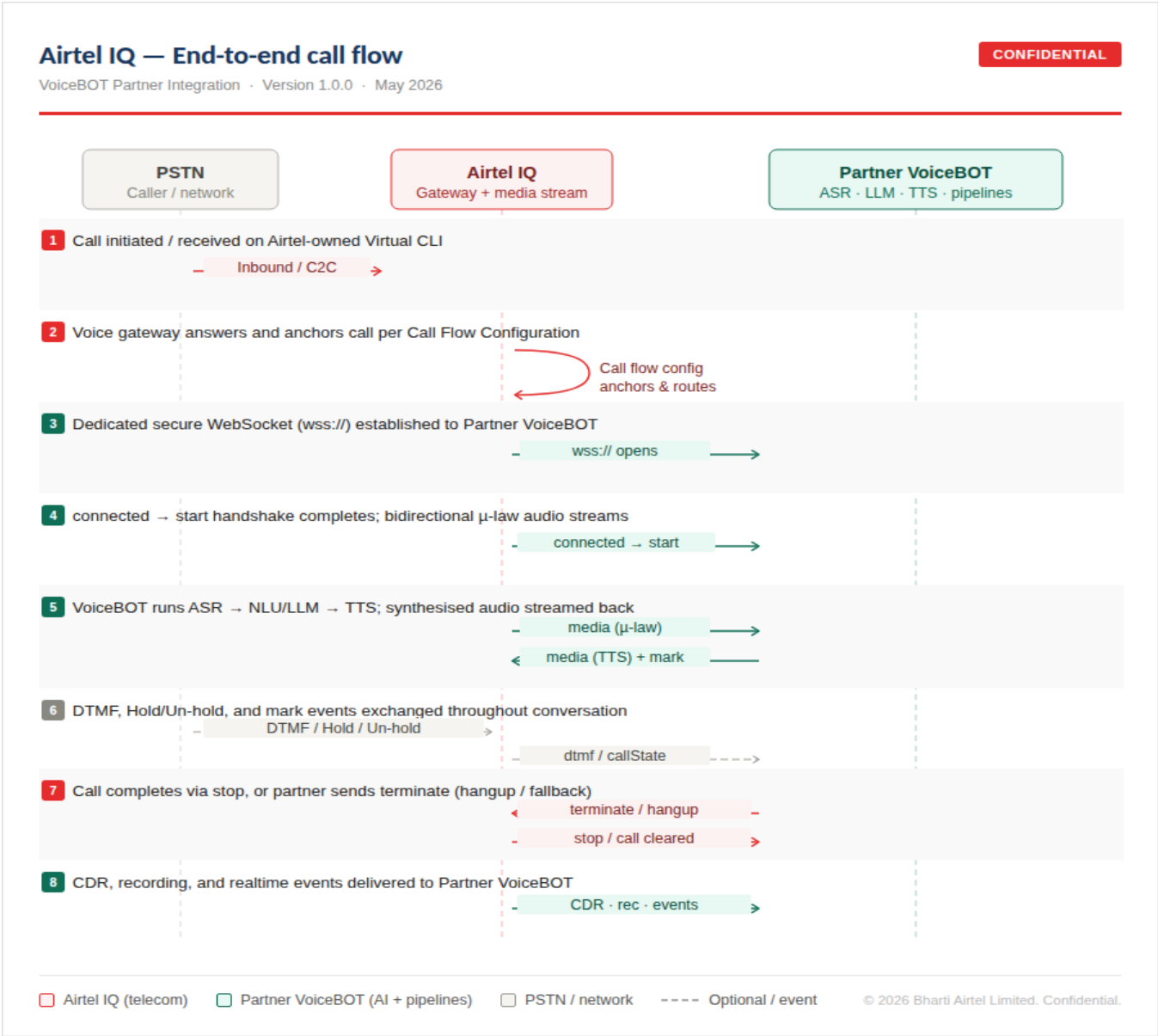
Airtel IQ uses a WebSocket Secure (WSS) media streaming contract to bridge real-time bidirectional audio between the telecom network and the Partner VoiceBOT Platform. This contract is the primary integration interface for Voice AI partners.

The Partner must expose a WSS endpoint satisfying all prerequisites (see Appendix 8.1). Once registered, Airtel IQ's Call Flow Configuration automatically opens a dedicated WebSocket connection per active call, streams μ -law 8 kHz audio in both directions, and exchanges JSON control messages throughout the call lifecycle.

4.4 End-to-End Call Flow

The following sequence describes the complete lifecycle of a voice call through the integrated platform:

- 1. Call is initiated or received by Airtel IQ on an Airtel-owned Virtual CLI.
- 2. Airtel IQ Voice Gateway answers and anchors the call as per Airtel-controlled Call Flow Configuration.
- 3. A dedicated secure WebSocket (wss://) is established to the Partner VoiceBOT.
- 4. connected → start handshake completes; bidirectional μ-law audio streams in real time.
- 5. Partner VoiceBOT runs ASR → NLU/LLM → TTS and streams synthesized audio back via media frames.
- 6. DTMF, Hold/Un-hold, and mark events are exchanged as needed throughout the conversation.
- 7. Call completes via stop, or partner sends termination (hangup / fallback to agent).
- 8. CDR, recording, and real-time events are delivered to the partner/customer via the agreed pipelines.



5. Partner Onboarding & Certification Checklist

The following checklist outlines the stages required to onboard and certify a new VoiceBOT Partner integration with Airtel IQ. Each stage must be completed and signed off before progressing to production go-live.

#	Checklist Item	Description
1	Architecture Review & Use-Case Sign-off	HLD walkthrough, ownership confirmation, traffic profile.
2	WebSocket & Audio Format Validation	wss:// handshake, connected/start contract, μ -law 8 kHz / 160-byte frame verification.
3	Functional Tests	Inbound, Outbound, C2C, Campaign, DTMF, Hold/Unhold, Mark, Clear, Terminate (Hangup & Fallback).
4	Load & Concurrency Testing	Peak CPS and concurrent calls; per-call dedicated WS connection validation.
5	Fallback & Failure Scenarios	Bot timeout, low confidence, WS disconnect, TTS failure → agent fallback.
6	CDR / Recording / Realtime Events Validation	SFTP push, Realtime CDR push, Realtime Events push, Recording pull/push.
7	Security & Compliance Approval	TLS, auth, data residency, PII handling.
8	Production Go-Live	Cutover plan, monitoring, on-call, rollback.

6. Security & Compliance

Area	Requirement
Transport	wss:// with TLS 1.2+ is mandatory in production. Plain WS (ws://) is not accepted.
Authentication	Bearer token / mTLS / signed parameters as agreed in the integration contract.
PII Handling	Caller/called numbers and custom parameters must be handled per applicable data protection regulations.
Recording Consent & Retention	Customer-owned policy; Airtel delivers; partner/customer retains per policy.
Audit	All API and WS sessions are logged on to Airtel IQ for audit and troubleshooting.

Technical References & API Specifications

7. Telephony Integration

In addition to the real-time WebSocket Media Streams contract, Airtel IQ exposes a set of HTTPS REST APIs for call orchestration, recording download, and post-call reporting. All Voice APIs require HMAC authentication.

7.1 HMAC Authentication

Airtel IQ Voice APIs are authenticated using HMAC-SHA256. HMAC ensures that only authorized callers can invoke the APIs, protect request integrity, and mitigate replay attacks via a time-bound X-Date header.

7.1.1 Key Elements

Element	Description
AppId	Your unique application ID, which identifies your client.
APIKey	Secret key shared between your client and the Airtel IQ server. Used to generate the HMAC signature; must be kept confidential.
X-Date	Timestamp of the request in RFC 1123 UTC format (e.g. "Mon, 06 Oct 2025 12:18:04 GMT"). Prevents replay attacks by limiting validity.
Digest	Base64-encoded SHA-256 hash of the request body, ensuring data integrity. Sent as the Digest header (value: SHA-256=<base64>).
Authorization	The final HMAC key is included in the Authorization header to authenticate the request.

7.1.2 Steps to Generate the HMAC Key

Step	Description
1	Generate the SHA-256 hash of the request body.
2	Encode the SHA-256 hash into a Base64 string to create the Digest.
3	Create the signatureRawData using the X-Date and Digest headers.
4	Generate the HMAC-SHA256 signature using the APIKey and signatureRawData.
5	Encode the generated signature into a Base64 string.
6	Generate the final HMAC key using AppId, X-Date, Digest, and the encoded signature.
7	Pass the HMAC key in the Authorization header of your API request.
8	Include the X-Date and Digest headers in the API request.

7.1.3 Sample (JavaScript) — HMAC Key Generation

```
var AppId = "<APP ID>";
var APIKey = "<API KEY>";
var requestTimeStamp = new Date().toUTCString();

var ssh = CryptoJS.SHA256(pm.request.body.toString());
var requestContentBase64String = CryptoJS.enc.Base64.stringify(ssh);
var digestvalue = "SHA-256=" + requestContentBase64String;

var signatureRawData = "x-date: " + requestTimeStamp + "\n" +
    "digest: " + digestvalue;
```

```

var signature      = CryptoJS.HmacSHA256(signatureRawData, APIKey);
var encodesignature = CryptoJS.enc.Base64.stringify(signature);

var hmacKey = "hmac username=" + JSON.stringify(AppId) +
              ", algorithm="    + JSON.stringify("hmac-sha256") +
              ", headers="      + JSON.stringify("x-date digest") +
              ", signature="     + JSON.stringify(encodesignature);

```

7.1.4 Required Headers on Every API Request

Header	Value
Authorization	<generated hmacKey>
Digest	SHA-256=<Base64(SHA-256(body))>
X-Date	RFC 1123 timestamp — e.g. "Mon, 06 Oct 2025 12:18:04 GMT"
Content-Type	application/json

NOTE: X-Date must use the exact RFC 1123 format: "Day, DD Mon YYYY HH:MM:SS GMT" (English abbreviations, 24-hour time, always GMT). A mismatched format or expired timestamp will result in HTTP 401 "HMAC signature does not match".

7.2 Click-to-Call API

The Click-to-Call API connects two participants on a call and optionally records the conversation. Airtel IQ first dials the first participant; once they answer, a call is initiated to the second participant. Both legs are bridged when the second party answers. Real-time call events are pushed to the partner's REST callback URL, and the final CDR (including the recording URL) is delivered when the call ends.

7.2.1 Endpoint

Property	Value
Method	POST
URL	/gateway/airtel-xchange/v2/adv/ob-click-to-call
Authorization	HMAC (see Appendix 1.1)
Content-Type	application/json

7.2.2 Request Headers

Header	Value
Authorization	Generated hmacKey
Digest	Digest value (SHA-256=<base64>)
X-Date	RFC 1123 standard date format
Content-Type	application/json

7.2.3 Request Body Parameters

Key	Type	M/O	Definition	Example
from	Integer/String	Mandatory	First number to be dialed.	"+ 919820XXXX"
to	List<Integer/String>	Mandatory	List of numbers to be dialed.	"+ 919820XXXX"
caller_id	String	Mandatory	Caller ID displayed to both parties.	"+ 91140XXXX"

Key	Type	M/O	Definition	Example
to_caller_id	String	Optional	Caller ID displayed to the callee. Overwrites caller_id for the callee leg.	—
record	Boolean	Optional	Set true to record the call. Defaults to false.	true / false
early_media	Boolean	Optional	Set true to enable early media before the call is connected.	true
callbacks	JSON Array	Optional	Overrides predefined CDR/event callbacks. Each entry: { event_type, notify_url, method, headers }.	—
retry	Integer/JSON	Optional	Retries on connection failure. Default 0, max 3.	{"count":2,"status":[...]}
metadata	JSON	Optional	Up to 3 client custom fields, passed over WebSocket and included in CDR.	—
call_flow_id	String	Mandatory	Unique Call Flow ID.	59487d0b-...

7.2.4 Callback URL Parameters

Name	Type	Description	Required
eventType	String	Event type to receive callbacks. One of: ALL, CALL, MEDIA, DTMF, RECORD, CDR, API, SUBMITTED, DELIVERED, ERROR.	Yes
headers	Object	Headers to send with the callback URL.	No
method	String	The HTTP method is supported by the callback URL. POST by default.	Yes
notifyURL	String	URL of the partner web service to be notified. Domains must be whitelisted by the Airtel Security Team. SSL certificate required for HTTPS endpoints.	Yes

7.2.5 Realtime Events

Airtel IQ posts real-time events to the callback URL specified in the request. Subscriptions may be for ALL types or a specific subset. Supported event types:

CalledNumber, Ringing, NotReachable, NoAnswer, Busy, Answer, RecordingStarted, PlaybackFinished, ChannelHold, ChannelUnhold, Matched, NotMatched, Timeout, Streaminfo, StreamError, terminate, Disconnected, RecordingFinished

Sample Event Payload

```
{
  "eventType": "CALL",
  "participantAddress": "{{ participantAddress }}",
  "event": "Removed",
  "time": "2020-02-26T12:19:46.399+0000",
  "metaData": {
    "causeCode": "16",
    "sipCode": "699",
    "causeCodeDescription": "Normal call clearing"
  }
}
```

NOTE: Airtel IQ expects an HTTP 200 OK response from your callback endpoint for every event.

7.3 Download Call Recording API

Download the call recording associated with a completed call.

7.3.1 Endpoint

Property	Value
Method	GET
URL	https://openapi.airtel.in/gateway/airtel-xchange/callv2/recording
Authorization	HMAC (see Appendix 1.1)
Content-Type	application/json

7.3.2 Request Headers

Header	Value
Authorization	Generated hmacKey
Digest	Digest value (SHA-256=<base64>)
X-Date	RFC 1123 standard date format
Content-Type	application/json

7.3.3 Request Body Parameters

Key	Type	M/O	Definition
token	String	Mandatory	Unique token received in the CDR "RecordingURL" used to download the recording.

7.3.4 Callback URL Parameters

The Download Recording API returns the recording file directly to the response body as a binary stream. The response is synchronous — no callback is used.

Sample Request (curl)

```
curl --location 'https://openapi.airtel.in/gateway/airtel-xchange/callv2/recording?token=<TOKEN>' \
  --header 'Content-Type: application/json' \
  --header 'Authorization: hmac username=<APP-ID>, algorithm="hmac-sha256",' \
  --header 'headers="x-date digest", signature="Eo5Sv28AcA4Yeyo6..."' \
  --header 'Digest: SHA-256=qEvowXa2ZG+aAinaerizzoV9MMw8sLXwc0WEY0ciuoU=' \
  --header 'X-Date: Tue, 07 Oct 2025 17:47:46 GMT'
```

7.4 Post Call / Campaign Reporting

Two delivery modes are supported for post-call CDR reporting:

7.4.1 Real-time CDR (Webhook)

If real-time CDRs are required, share a Webhook URL with Airtel IQ. After each call, the CDR is HTTP POSTed to the shared endpoint in the format shown below.

Sample CDR Payload

```
{
  "Overall_Call_Status":    "Answered",
  "Caller_ID":              "${caller_id}",
  "callType":               "OUTBOUND",
```

```
{
  "duration": 17484,
  "Billable_Duration": "00:09",
  "conversationDuration": 932,
  "startTime": 1762164125867,
  "endTime": 1762164143351,
  "Session_ID": "1f5040db-1fbf-4604-aeae-6b0d1445a6d4",
  "Hangup_Cause": "Destination",
  "destinationNumber": "${destination_mobile_number}",
  "metaData": {
    "Branch_code": "${branch_code}",
    "campaignId": "1f5040db-1fbf-4604-aeae-6b0d14aaaa"
  },
  "RecordingURL": "https://openapi.airtel.in/gateway/.../recording/download?token=..."
}
```

Expected Response from Partner Endpoint

```
{
  "status": "success"
}
```

NOTE: Required CDR fields should be confirmed with the customer during onboarding — the sample above is illustrative.

7.4.2 Scheduled SFTP Delivery

Airtel IQ can deliver all generated CDRs in a CSV file over SFTP at a specified time of day. SFTP credentials and destination paths are required from the partner/customer during onboarding.

RECOMMENDATION: Minimum SFTP file interval is 2 hours. End-of-day SFTP delivery is strongly recommended to evenly distribute system load and prevent performance degradation.

8. Media Streaming based Voice Agent Platform Integration

Every message — whether sent by Airtel IQ or by your server — is a JSON text frame with at minimum an event field that identifies the message type. This appendix details the complete WebSocket contract for media streaming integration.

8.1 Prerequisites & WSS Endpoint Requirements

Before onboarding, the partner must expose a WebSocket Secure (WSS) endpoint that satisfies all the requirements listed below. The endpoint URL is registered against the partner account in the Airtel IQ Partner Portal and is referenced by the Airtel-controlled Call Flow Configuration.

8.1.1 Endpoint URL Format

wss://<your-domain>/<path>

Example: wss://company.example.com/airtelqiqstream

8.1.2 Requirements Checklist

Requirement	Detail
Protocol	WSS (TLS 1.2 or higher). Plain WS (ws://) is NOT accepted in production.
TLS Certificate	Valid, CA-signed certificate. Self-signed certificates are rejected.
Port	443 (recommended).
URL Registration	Your WSS URL must be registered in the Airtel IQ Partner Portal.
Concurrent Connections	Each active call opens exactly one WebSocket connection. Plan capacity accordingly.
Message Format	All messages are JSON-encoded UTF-8 text frames. Binary frames are not used.
Authentication	Optional authentication may be enabled (Bearer token / mTLS / signed query parameter — agreed bilaterally during onboarding).
Firewall Configuration	Allow Airtel's Public IP ranges (shared during onboarding).

NOTE: Airtel IQ will not retry a failed WebSocket handshake. This impacts caller experience — the call will be dropped with or without announcement. Your endpoint must be highly available.

8.2 Airtel IQ WebSocket Contracts

8.2.1 Common Message Envelope

Every message sent by either party is a JSON text frame containing at minimum the following envelope fields:

Field	Description
event	String. Identifies the message type. One of: connected, start, media, stop, dtmf, callState, mark, clear, terminate.
streamSid	String. Unique identifier for this stream session. Present all messages after the connected event.

8.2.2 WebSocket Messages from Airtel IQ

Airtel IQ sends the following message types to your WebSocket server during an active call stream:

Message	Description
connected	Handshake confirmation; first message ever received.
start	Call/stream metadata; sent immediately after connected.
media	Raw audio frames (inbound and/or outbound).
dtmf	DTMF key-press events from the caller (bidirectional streams only).
stop	Stream has ended or call has been disconnected.
call State	Call placed on hold or un-hold.
mark	Playback completion acknowledgement (bidirectional streams only).

8.2.2.1 connected

The connected message is the very first message sent by Airtel IQ upon successfully opening the WebSocket connection. It describes the protocol version your server can expect for the lifetime of this connection.

Field	Type	Description
event	string	Always "connected". Identifies this message type.
protocol	string	Protocol identifier. Value is always "Call".
version	string	Semantic version of the Airtel IQ streaming protocol. Current value: "1.0.0".

Example JSON

```
{
  "event": "connected",
  "protocol": "Call",
  "version": "1.0.0"
}
```

8.2.2.2 start

The start message follows immediately after connected. It is sent exactly once and carries all call and stream metadata your Voice Agent needs to process the session.

Field	Type	Description
event	string	Always "start".
start.streamSid	string	The unique identifier of the stream.
start.accountSid	string	Airtel IQ account identifier assigned to your organisation.
start.callSid	string	Unique identifier for the originating call leg.
start.mediaFormat.encoding	string	Audio codec. Always "audio/x-mulaw" (G.711 μ -law).
start.mediaFormat.sampleRate	string	Sample rate in Hz. Always 8000.
start.mediaFormat.channels	string	Number of audio channels. Always 1 (mono).
start.callerNumber	string	The caller number.
start.calledNumber	string	The called number.
start.customParameters	string	Key-value pairs defined when the stream was configured. Use to pass call context (e.g. language, customer ID, intent).
streamSid	string	The unique identifier of the stream (top-level copy).

Example JSON

```
{
  "event": "start",
  "start": {
    "streamSid": "2ee80e99-af61-442c-a10f-62fceb32fa1a_AirtelDigital_aa1144b8-d5a2-4e78-94b3-5d158664971c",
    "accountSid": "Voice_Test",
    "callSid": "2ee80e99-af61-442c-a10f-62fceb32fa1a",
    "mediaFormat": {
      "encoding": "ulaw",
      "sampleRate": "8000",
      "channels": "1"
    },
    "callerNumber": "7709977763",
    "calledNumber": "7709977764",
    "customParams": "{\"caseId\": \"xyz\"}"
  },
  "streamSid": "2ee80e99-af61-442c-a10f-62fceb32fa1a_AirtelDigital_aa1144b8-d5a2-4e78-94b3-5d158664971c"
}
```

8.2.2.3 media

The media message carries raw audio payload. Airtel IQ streams audio continuously from the moment the stream starts until the call ends. Each media message contains a single audio chunk.

Field	Type	Description
event	string	Always "media".
streamSid	string	Identifies the stream session this audio belongs to.
media.payload	string	Base64-encoded raw μ -law 8000 Hz mono audio. Decode to PCM before processing. Do NOT include any file-header bytes when sending audio back.

NOTE: The payload contains raw audio bytes with NO file-type header. Do not prepend WAV/AU headers when sending audio back — doing so will cause distorted playback. For μ -law, each WebSocket message carries an audio payload of 160 bytes.

Example JSON

```
{
  "event": "media",
  "streamSid": "2ee80e99-af61-442c-a10f-62fceb32fa1a_AirtelDigital_aa1144b8-d5a2-4e78-94b3-5d158664971c",
  "media": {
    "payload": "no+JhoaJj pzS8vLy8vP..."
  }
}
```

8.2.2.4 dtmf

The dtmf message is sent when the caller presses a dial-pad key. Only available on bidirectional streams; can be used to implement IVR-style fallback within your Voice Agent.

Field	Type	Description
event	string	Always "dtmf".
streamSid	string	Identifies the stream session.
dtmf.digit	string	The digit pressed. One of: 0–9, *, #.

Example JSON

```
{
  "event": "dtmf",
  "streamSid": "2ee80e99-af61-442c-a10f-62fceb32fa1a_AirtelDigital_aa1144b8-d5a2-4e78-94b3-5d158664971c",
  "dtmf": {
    "digit": "1"
  }
}
```

2.2.2.5 stop

The stop message is the final message Airtel IQ sends. It signals that the stream has ended — either because the call was disconnected, or because the stream was explicitly terminated using a terminate message. Your server should close any associated resources upon receiving this event.

Field	Type	Description
event	string	Always "stop".
stop.accountSid	string	Airtel IQ account identifier.
stop.callSid	string	Identifier of the call leg that generated this stream.
streamSid	string	Identifies the stream session that has ended.

Example JSON

```
{
  "event": "stop",
  "stop": {
    "streamSid": "32275366-716b-4f16-9fb9-b1947708bd4f_AirtelDigital_8e62839b-8f92-4462-bbcb-52cc42e50486",
    "accountSid": "Voice_Test",
    "callSid": "32275366-716b-4f16-9fb9-b1947708bd4f"
  },
  "streamSid": "32275366-716b-4f16-9fb9-b1947708bd4f_AirtelDigital_8e62839b-8f92-4462-bbcb-52cc42e50486"
}
```

NOTE: After sending stop, Airtel IQ will close the WebSocket connection with normal closure code 1000. Your server need not send any response message.

8.2.2.6 call state

Airtel IQ sends a callState message when the user has pressed Hold or Un-hold on the call.

Field	Type	Description
event	string	Always "callState".
streamSid	string	Identifies the stream session.
call.state	string	One of "hold" or "unhold".

Example JSON — hold

```
{
  "event": "callState",
  "streamSid": "873cf3f4-2c6d-4eba-8e48-9928465eadc9_AirtelDigital_36edbe1a-7b38-4909-8ba0-0c2e5c231146",
  "call": {
    "state": "hold"
  }
}
```

Example JSON — unhold

```
{
  "event": "callState",
  "streamSid": "873cf3f4-2c6d-4eba-8e48-9928465eadc9_AirtelDigital_36edbe1a-7b38-4909-8ba0-0c2e5c231146",
  "call": {
    "state": "unhold"
  }
}
```

8.2.2.7 mark

After you send media messages to Airtel IQ, you can send a mark message. When media playback is complete, Airtel IQ sends this mark message back to your server with the same mark.name field. Use it to synchronize your agent's dialogue turns.

Field	Type	Description
event	string	Always "mark".
streamSid	string	Identifies the stream session.
mark.name	string	The label you originally sent in your mark message. Use this to match completions to your sent audio segments.

Example JSON

```
{
  "event": "mark",
  "streamSid": "62fceb32fala_AirtelDigital_aa1144b8-d5a2-4e78-94b3-5d158664971c",
  "mark": {
    "name": "mark-for-some-announcement"
  }
}
```

8.2.3 WebSocket Messages to Airtel IQ

For bidirectional streams, your Voice Agent server can send the following message types back to Airtel IQ to inject audio or control the stream:

Message	Description
media	Inject agent audio into the call.
mark	Set a playback-completion notification.
clear	Flush the pending audio buffer (barge-in / interruption).
terminate	Hang up the caller or fall back to a human agent.

NOTE: Audio *MUST* be μ -law encoded at 8000 Hz, mono, base64-encoded, with NO file-type headers. Any other format will result in garbled audio or silent playback.

8.2.3.1 media

Send a media message to inject synthesised or pre-recorded audio into the active call. Audio chunks are buffered by Airtel IQ and played sequentially. To interrupt buffered audio, send a clear message first.

Field	Type	Description
event	string	Always "media".
streamSid	string	The streamSid received in the start message. Must match exactly.
media.payload	string	Base64-encoded raw μ -law 8000 Hz mono audio bytes. Use smaller chunks (20–40 ms) to reduce perceived latency.

Example JSON

```
{
  "event": "media",
  "streamSid": "2ee80e99-af61-442c-a10f-62fceb32fala_AirtelDigital_aa1144b8-d5a2-4e78-94b3-5d158664971c",
  "media": {
    "payload": "no+JhoaJj pzS8vLy8vP..."
  }
}
```

8.2.3.2 mark

Send a mark message after sending media messages to get notified when the audio you have sent has completed playback. Airtel IQ sends back a mark message with the same mark.name when audio preceding your mark has been played.

Field	Type	Description
event	string	Always "mark".
streamSid	string	The streamSid received in the start message.
mark.name	string	A label of your choice (e.g. "turn-1-end", "greeting-done"). Must be unique within the stream.

Example JSON

```
{
  "event": "mark",
  "streamSid": "62fceb32fala_AirtelDigital_aa1144b8-d5a2-4e78-94b3-5d158664971c",
  "mark": {
    "name": "greeting-complete"
  }
}
```

8.2.3.3 clear

Send a clear message to interrupt audio that has already been sent in various media messages. This empties all buffered audio and causes remaining mark messages to be sent back to your WebSocket server.

Field	Type	Description
event	string	Always "clear".
streamSid	string	The streamSid received in the start message. Identifies which stream's buffer to flush.

Example JSON

```
{
  "event": "clear",
  "streamSid": "62fceb32fala_AirtelDigital_aa1144b8-d5a2-4e78-94b3-5d158664971c"
}
```

8.2.3.4 terminate

Send a terminate message to end the call or fall back to a human agent. Two actions are supported: Hangup (hang up the caller and terminate streams) and Fallback (connect the caller to a human agent).

Field	Type	Description
event	string	Always "terminate".
streamSid	string	The streamSid received in the start message.
reason.code	integer	Reason code for termination. Determines the action taken (see Terminate Reason Codes below).
reason.text	string	Informative message with appropriate reasoning for termination/fallback.
reason.fallbackNumber	string	E.164 compliant number where the fallback call has to be initiated (when code = 3).

Example JSON

```
{
  "event": "terminate",
  "streamSid": "2ee80e99-af61-442c-a10f-62fceb32fa1a_Air_aa1144b8-d5a2-4e78-94b3-5d158664971c",
  "reason": {
    "code": 1,
    "text": "normal clearing",
    "fallbackNumber": "+918219721982"
  }
}
```

Terminate Reason Codes

reason.code	reason.text	Action
1	Informative message with appropriate reasoning for termination.	Hangup the Caller
2	Informative message with appropriate reasoning for fallback.	Fallback to Agent
3	Set reason.fallbackNumber to an E.164 compliant number where fallback call has to be initiated.	Fallback to Particular DID

8.2.4 Sample Message Flow

The sequence below illustrates a complete bidirectional stream lifecycle for a Voice Agent turn:

Airtel IQ → Your Server	Direction	Your Server → Airtel IQ
connected	→	
start	→	
media	→	
media	→	
	←	media (agent TTS audio)
	←	mark ("greeting-done")
dtmf (caller pressed key)	→	
mark ("greeting-done")	→	
media	→	
	←	clear (barge-in)
	←	media (new response audio)
	←	mark ("response-1-end")
mark ("response-1-end")	→	

Airtel IQ → Your Server	Direction	Your Server → Airtel IQ
stop	→	

End